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# Fluid dynamics of local martian magma oceans

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**Abstract**

The evolution of a melt region produced by a large impact during Mars formation is addressed. While some impact induced melt is redistributed during crater excavation, sufficiently large impacts (much larger than basin forming impacts) generate an intact melt region which is retained beneath the excavation zone, i.e., a local magma ocean. Local magma ocean evolution depends on the effective rheology controlling large scale deformation of the solid part of the planet, mechanism of crystallization, and melt region size. Within the uncertainties of various parameters, two scenarios are possible. For sufficiently weak rheology or large melt region size, evolution is characterized by rapid extrusion and formation of a global magma ocean. For sufficiently strong rheology or small melt region size, in situ crystallization to a partially molten solid state occurs prior to isostatic adjustment. Subsequent to in situ crystallization, local magma ocean evolution depends on melt region size and efficiency of lateral redistribution compared to bulk conductive cooling. For large melt regions, lateral spreading occurs via plastic deformation and results in an asymmetric, global, partial melt layer. For small melt region size, viscous spreading viscous can result in bulk cooling below the solidus prior to formation of a global layer. A hypothesis for the origin of the hemispherical crustal dichotomy and Tharsis rise is suggested. The dichotomy is associated with a global partial melt layer produced by evolution of a large, local magma ocean. After dichotomy formation, evolution of a second, smaller, local magma ocean is related to Tharsis development.
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## 1. Introduction

Geological and geophysical observations suggest that early Mars was quite dynamic with several global scale processes occurring early in planetary history including rapid core formation and crustal differentiation (e.g., Lee and Halliday, 1997), a transient core dynamo (Stevenson, 2001), and early development of both the hemispherical crustal dichotomy (Zuber, 2001; Frey et al., 2002; Nimmo and Tanaka, 2005) and Tharsis (Banerdt and Golombek, 2000; Phillips et al., 2001; Johnson and Phillips, 2005; Solomon et al., 2005). One crucial aspect of early martian history which is key to explaining these observations is the inevitability of giant impacts during planetary formation (e.g., Agnor et al., 1999) and the implications of such impacts for the initial thermal and compositional state of the planet. Giant impacts are much larger than those which formed the Hellas and Utopia basins (Hood et al., 2003; Mohit and Arkani-Hamed, 2004) and are expected based on improved models of late stage terrestrial planet accretion (Wetherill, 1990; Chambers and Wetherill, 1998; Agnor et al., 1999). Such impacts deposit sufficient energy via the shock wave to melt a large fraction of proto-Mars.

An initially hot and extensively molten Mars has been postulated based on geochemical evidence for early core–mantle–crust differentiation (Chen and Wasserburg, 1986; Righter and Drake, 1996; Lee and Halliday, 1997; Brandon et al., 2000; Halliday et al., 2001; Kleine et al., 2004). Both Rb/Sr and Sm/Nd data for shergottites define a 4.5 Gyr isochron (Borg et al., 1997, 2003) providing evidence for early mantle melting. The $^{182}$Hf–$^{182}$W chronometer indicates core formation during the lifetime of $^{182}$Hf. A recent reanalysis of the chondritic $^{182}$W/$^{184}$W ratio (Kleine et al., 2004) implies that the end of core formation is constrained to 7–15 Myr [the original estimate (Lee and Halliday, 1997) was ∼30 Myr]. Furthermore, correlation between $^{182}$W anomalies and $^{146}$Sm–$^{142}$Nd

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data (Harper et al., 1995) for the same samples is consistent with approximately simultaneous metal–silicate separation and crustal differentiation (Lee and Halliday, 1997). Finally, correlation of $^{187}$Re–$^{187}$Os data with both $^{142}$Nd and $^{182}$W anomalies requires Re/Os fractionation when both $^{146}$Sm and $^{182}$Hf were extant (Brandon et al., 2000). A martian magma ocean facilitates effective metal–silicate separation (Stevenson, 1990) although other mechanisms can contribute to percolative core formation (Bruhn et al., 2000; Rushmer et al., 2000; Hustoft and Kohlstedt, 2006). A magma ocean also has important implications for initial compositional differentiation, compositional convection, magnetic field generation (Elkins-Tanton et al., 2003) and may result in a stably stratified mantle which would tend to delay or even suppress subsequent thermal convection (Zaranek and Parmentier, 2004).

Development of a global magma ocean from an impact induced local magma ocean requires rapid redistribution of the buoyant melt region prior to any significant solidification (Tonks and Melosh, 1992, 1993). If cooling and crystallization is fast compared to melt redistribution, the isostatic adjustment driving force and melt region viscosity decrease and increase, respectively. In this case, restoration of a globally symmetric mass distribution can be difficult. Large scale thermal and compositional heterogeneity associated with local magma ocean evolution can shape the initial crustal distribution and early mantle dynamics. The purpose of the study is to present a simple fluid dynamical model of local magma ocean evolution which can be subject to further hypothesis testing based on observations related to the timing of formation of crustal structure, cessation of the dynamo, and other geological and geophysical data. Understanding the evolution of impact induced local magma oceans can provide a fundamental framework for understanding early martian evolution. A hypothesis is suggested in which hemispheric dichotomy formation and Tharsis development, which occurred early in martian history, are related to the evolution of two local magma oceans.

The study is organized as follows. Section 2 is a review of the physics of impact melting and local magma ocean formation (Tonks and Melosh, 1992, 1993). Section 3 is an estimation of isostatic adjustment and crystallization timescales. The question of globally symmetric redistribution of melt versus in situ solidification to a partially molten state is addressed and two end-member scenarios are identified. Section 4 is a study of the evolution of a partially molten, local magma ocean subsequent to crystallization. Timescales are estimated for convective cooling, differentiation, and isostatic adjustment. In Section 5, a hypothesis for the origin of the hemispherical dichotomy and Tharsis is suggested and several additional questions raised by the study regarding early martian evolution are discussed.

## 2. Impact melting

### *2.1. What was Mars' thermal state during early accretion prior to large impacts?*

While the specific gravitational binding energy, $3/5GM/R_p$, where $G$ is the gravitational constant, $M$ is the planet mass, and $R_p$ is the planet radius, is sufficient to melt the entire planet, energy retention efficiency depends on impact heating of the interior and impact mechanisms which advect internal energy to the surface (e.g., excavation and isostatic rebound) where it can be radiated away (Safronov, 1978; Kaula, 1979). Since impact energy burial and excavation depend on size and frequency of impacts, retention efficiency is a function of the planetesimal size distribution and accretion timescale. The size correlation between Mars and planetary embryos (Wetherill and Stewart, 1989; Weidenschilling et al., 1997; Kokubo and Ida, 1996, 1998, 2000) suggests that Mars could be a remnant of the runaway growth stage (Chambers and Wetherill, 1998) which avoided several embryo–embryo scale collisions characteristic of the final stage of planetary formation (Wetherill, 1990; Chambers and Wetherill, 1998; Agnor et al., 1999). Runaway growth occurs on a timescale of ∼1 Myr and embryos are formed from a mass distribution with most mass in the smallest bodies (Kokubo and Ida, 1996, 1998, 2000). A recent study of Mars thermal state during accretion with a new impact heating formulation, inclusion of metal–silicate separation, an impactor size distribution consistent with improved models of runaway growth of planetary embryos (Kokubo and Ida, 1996, 1998, 2000) and short accretion time of ∼1 Myr indicate a cold martian interior and onset of melting only when the radius of proto-Mars is $\sim 0.8R_p$ (Senshu et al., 2002). Planetesimal heating by electromagnetic induction (Herbert et al., 1991), collisions (Rubin, 2003, 2004), and decay of the short-lived radioisotopes $^{26}$Al and $^{60}$Fe (Urey, 1955; Grimm and McSween, 1993; Shukolyukov and Lugmair, 1993; Mostefaoui et al., 2005) can contribute to planetary melting during formation. Heating by short lived radioisotopes is sensitive to planetesimal size, the initial radioisotope concentration, and planetesimal accretion timescale (Merk et al., 2002). If the thermal conduction timescale based on the planetesimal size is smaller than the radioisotope half-life, significant heating is unlikely. Accounting for accretion enhances cooling of small bodies (Merk et al., 2002), while significant melting is likely for larger planetesimals. No models of Mars' thermal state during accretion have included the effect of planetesimal heating. While rapid accretion (∼1 Myr) alone is not a guarantee of complete melting during Mars formation, a mostly solid state is difficult to reconcile with early core formation which probably requires some degree of partial melting. In this study, the focus is on large impacts for which significant melting of a large fraction of the planet is unavoidable (Tonks and Melosh, 1992, 1993).

### *2.2. How much melt is produced by an impact?*

Runaway growth models predict a gap between the planetary embryo mass and the upper bound of the continuous planetesimal mass distribution spanning ∼1 mass decade (Wetherill, 1990; Kokubo and Ida, 1996, 1998, 2000). Thus, assuming that Mars is indeed a remnant planetary embryo (Chambers and Wetherill, 1998) it is likely that a fraction of Mars mass was delivered in the form of impacting bodies with masses ∼0.1–1 times the mass of the Moon corresponding to radii

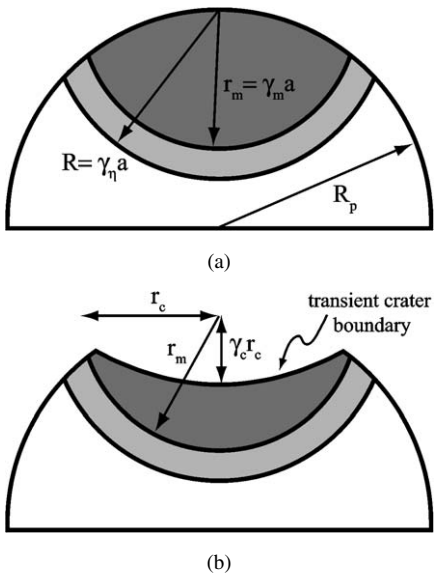


Fig. 1. (a) Hemispherical model of impact melt distribution. The radii of the complete melt region (dark gray) and region with melt fraction $\phi > \phi_{cr}$ (light gray) are $r_m$ and $R$, respectively. The impactor radius is $a$ and the proportionality coefficients $\gamma_m = 1.3$ and $\gamma_\eta = 1.7$. (b) Melt distribution subsequent to the excavation stage of a large impact. For sufficiently large impacts, a fraction of impact melt is unexcavated by the cratering flow. The transient crater radius and depth to radius ratio are $r_c$ and $\gamma_c$, respectively. The dependence of the retained melt fraction $f$ on the ratio $r_m/r_c$ is given by Eq. (7).

~800–1700 km. However, the number of bodies at the upper end of the mass distribution is small and, while the effects of such impacts may be crucial to understanding the Mars' early state, they are also inherently controlled by the statistics of small numbers.

Large impacts generate a spherical shock wave which propagates into the planet. Near the impact site, there is an isobaric core where shock pressures are approximately uniform. Outside the isobaric core, shock pressures decay with distance. For a hemispherical model (Fig. 1) with vertical incidence and a target to projectile density ratio of 1, the radius of the region shocked to pressure $P$ or higher is (Tonks and Melosh, 1992, 1993)

$$r = 2^{1/3}\left(\frac{v_i}{v_i^P}\right)^{1/2} a, \tag{1}$$

where $a$ is the impactor radius, $v_i$ is the impact velocity, and

$$v_i^P = \frac{c}{s}\left[\left(1 + \frac{4sP}{\rho c^2}\right)^{1/2} - 1\right], \tag{2}$$

with material parameters $c$, $s$ and density $\rho$. For dunite at shock pressures $P \geqslant 73$ GPa, $c = 4.4$ km/s and $s = 1.5$ (Kieffer, 1977). The shock pressure corresponding to complete melting of dunite initially at a pressure of 1 bar and solidus temperature is $P_m = 115$ GPa (Tonks and Melosh, 1993) corresponding to $v_i^{P_m} = 7$ km/s. For impact velocity $v_i \sim 7$ km/s, the radius of the complete melt region $r_m = \gamma_m a$, where $\gamma_m = 2^{1/3} \approx 1.3$. Oblique impacts complicate this simple geometry, concentrating more energy downrange (Pierazzo and Melosh, 2000).

As the radius from the impact site increases beyond $r_m$, the melt fraction decreases. The radius of a region shocked to a melt fraction $\phi$ can be calculated from the Hugoniot in pressure–entropy space (Tonks and Melosh, 1993) along with the assumption that melt fraction scales linearly with entropy, $S$,

$$\phi = \frac{S - 2485\ \mathrm{J/kg\,K}}{750\ \mathrm{J/kg\,K}}. \tag{3}$$

The critical melt fraction $\phi_{cr} \sim 0.4$ corresponds to a rheological transition from low viscosity liquid to high viscosity partially molten solid (see A.3). This corresponds to an entropy of 2785 J/kg K and thus a shock pressure of ~70 GPa. With $P = 70$ GPa, $v_i^P = 4.6$ km/s and the ratio of the radius of the region with $\phi \geqslant \phi_{cr}$ to the impactor radius $\gamma_\eta = R/a \sim 1.7$ [Fig. 1(a)].

In the infinite half-space approximation ($r_m/R_p \ll 1$), the melt volume is simply $V_m = 2/3\pi r_m^3$. For larger impacts, planetary surface curvature is important and the melt volume (Tonks and Melosh, 1993),

$$V_m = \frac{1}{2} V_p \xi_m^3 (1 - 3/8\xi_m), \tag{4}$$

where $\xi_m = r_m/R_p$ is the ratio of the complete melt region radius to the planetary radius and $V_p = 4/3\pi R_p^3$ is the planetary volume. Likewise the volume of the region with $\phi \geqslant \phi_{cr}$ is

$$V = \frac{1}{2} V_p \xi^3 (1 - 3/8\xi), \tag{5}$$

where $\xi = R/R_p$.

### 2.3. What melt fraction is retained after excavation?

The material flow field behind the shock front results in excavation of a fraction of the melt region [Fig. 1(b)]. The geometric figure of the excavation zone can be approximated with the Maxwell $Z$-model (Maxwell, 1977),

$$r = r_c (1 - \cos\varphi)^{1/(z-2)}, \tag{6}$$

where $\varphi$ is the angle from vertical, $r_c$ is the radius of the transient crater (i.e., the excavated cavity prior to any floor rebound or rim collapse), $r$ the distance from the impact site, and $z = 3$ gives the best approximation to the cratering flow field. The retained melt fraction is the fraction of melt volume remaining after excavation $f = 1 - V_{m,e}/V_m$, where $V_{m,e}$ is the excavated melt volume,

$$f = \begin{cases} 3/4(r_m/r_c), & r_m < r_c, \\ 1 - 1/4(r_m/r_c)^{-3}, & r_m > r_c. \end{cases} \tag{7}$$

The transient crater radius is given by the $\pi$-scaling law for the gravity regime (Schmidt and Housen, 1987),

$$r_c = \frac{1}{2}\left(\frac{4}{3}\pi\right)^{1/3} C_D \left(3.22\frac{g}{v_i^2}\right)^{-\beta} a^{1-\beta}, \tag{8}$$

where $g$ is the gravitational acceleration, and $C_D = 1.6$ and $\beta = 0.22$ are constants (Melosh, 1989). For an impact velocity $v_i \sim 7$ km/s,

$$r_c \approx 300\ \mathrm{km}\left(\frac{a}{100\ \mathrm{km}}\right)^{0.78}. \tag{9}$$

Impact shock pressure magnitude is determined primarily by impactor velocity and target material properties while the shock pressure distribution depends mainly on impactor size. In the case of normal incidence, the shock region size scales with impactor size (O'Keefe and Ahrens, 1977) and, if melting occurs ($v_i > v_i^{P_m}$), the melt region radius $r_m \sim a$. Small incidence angles complicate melt region geometry and numerical simulations indicate that melt volume decreases with decreasing impact angle (Pierazzo and Melosh, 2000). The transient crater dimension, on the other hand, is sensitive to gravitational acceleration. In pi-scaling theory, the dimensionless transient crater radius scales with gravity scaled impactor size, $r_c/a \sim (ga/v_i^2)^{-\beta}$. Clearly, $\beta < 0$ since $\partial r_c/\partial v_i > 0$ and $\partial r_c/\partial g < 0$. Also, $\beta < 1$ since $\partial r_c/\partial a > 0$. It follows that $r_c$ increases slower than linearly with $a$, $r_c \sim a^{1-\beta}$. The experimental value for competent rock $\beta \sim 0.22$ (Schmidt and Housen, 1987; Melosh, 1989). The differential scaling $r_m \sim a$ and $r_c \sim a^{1-\beta}$ imply that as impactor size increases, the melt region radius increases faster than the crater radius. This result has been noted previously in the context of core formation (Tonks and Melosh, 1992), magma ocean formation (Tonks and Melosh, 1993), and terrestrial (Grieve and Cintala, 1992) and lunar (Cintala and Grieve, 1998) impacts. The retained melt fraction [Eq. (7)] increases with increasing $a$ and eventually a large volume of melt is retained.

### *2.4. What are the conditions for formation of an intact melt region?*

Sufficiently large impacts generate melt deep in a planet where it remains subsequent to crater excavation [Fig. 1(b)] forming an intact melt region. The evolution of such a melt pond on Mars is the subject of this study. While an exact criterion for intact melt region formation is not easily defined, Tonks and Melosh (1992) suggest: (i) transient crater collapse controlled by melt viscosity and not the flow of solids, and (ii) sufficient melt retained to form a pool at the transient crater bottom. These criteria are marginally satisfied if the melt region just reaches the bottom of the transient crater and are easily satisfied if the melt volume equals the transient crater volume. For a transient crater with a parabolic cross section (Melosh, 1989), radius $r_c$, and depth to radius ratio $\gamma_c$, the transient crater volume

$$V_c = \frac{\gamma_c}{2}\pi r_c^3. \tag{10}$$

Equating Eqs. (4) and (10) yields a critical retained melt fraction corresponding to the equal melt and crater volume criterion

$$f_{cr} = \frac{3}{4}\left(\frac{3}{4}\gamma_c\right)^{1/3}\left(1 - \frac{3}{8}\xi_m\right)^{-1/3}. \tag{11}$$

Typically, the transient crater depth to diameter ratio is ≈1/4–1/3 so $\gamma_c \approx 1/2$–$2/3$. To zeroth order in $\xi_m$ (accurate to within ∼20% for the largest $\xi_m \sim 1$), $f_{cr} \approx 0.5$–$0.6$ (Tonks and Melosh, 1992). The critical impactor radius satisfies

$$\gamma_m a_{cr} = \frac{4}{3} f_{cr} r_c(a_{cr}), \tag{12}$$

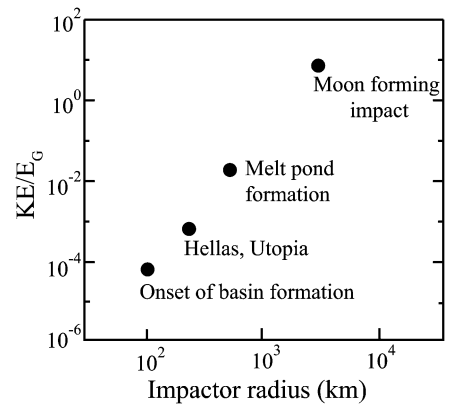


Fig. 2. Comparison of impactor kinetic energies scaled by the martian gravitational binding energy $E_G$ for different size events. Formation of a Hellas/Utopia size basin is based on a ∼200 km radius impactor. The energy of a Moon forming impact on the proto-Earth is based on an impactor to target mass ratio ∼0.1 and impact velocity $v_i \sim v_{\mathrm{esc}_{\mathrm{Earth}}}$.

where $r_c(a)$ is the crater radius scaling [Eq. (9)]. Thus, $a_{cr} \approx$ 700–1500 km. This estimate is an upper bound since only the volume of the complete melt region was considered and partial melt regions can contribute (Tonks and Melosh, 1993).

Magma pond forming impacts are significantly larger and more energetic than basin forming impacts. Basins such as Hellas and Utopia were formed by impactors with radii much smaller than $a_{cr}$. Estimates of the Hellas impactor radius range from 200–230 km (Mohit and Arkani-Hamed, 2004) to 230–340 km (Hood et al., 2003) a factor of 2–8 times smaller than $a_{cr}$. Since impactor kinetic energy scales with $\sim a^3$, the lower bound of melt region forming impact energies are a factor of 10–30 times larger than impact energies corresponding to the largest observed basins (Fig. 2). The lower bound on melt region size corresponds to $\xi_{\min} \sim \gamma_\eta a_{cr}/R_p \sim 0.35$. The upper bound $\xi_{\max}$ satisfies $\xi_{\max}^3 - 3/8\xi_{\max}^4 \sim 1$ (corresponding to $V/V_p \sim 1/2$) such that $\xi_{\max} \sim 1.2$. For larger $\xi$, the ratio of the melt volume to solid volume exceeds 1 and the geometry becomes more complicated due to the transition from a mostly solid planet with a localized melt region to a predominantly molten planet.

## 3. Does the local magma ocean form a global melt layer or crystallize in place?

The intact melt region generated by a giant impact can form a global magma ocean only if the ratio of the isostatic adjustment time to the melt cooling time is small (Tonks and Melosh, 1992, 1993). The crystallization time is a function of convective cooling of the melt and, in particular, the degree to which cold, partially solid material near the surface participates in the flow. Complete isostatic adjustment involves both radial relaxation as the buoyant melt region moves upward and a lateral component as melt is redistributed around the planet to a spherically symmetric state. The radial relaxation timescale is sensitive to large scale deformation of the solid part of the planet. Lateral relaxation involves deformation of the low viscosity melt region itself. Surface cooling and crust formation result in large

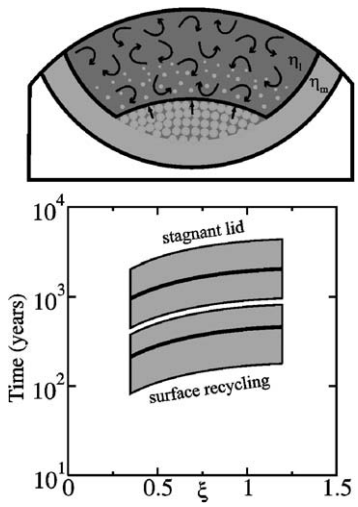


Fig. 3. (Top) Melt region crystallization. As the magma pond cools via convection, the temperature drops below the liquidus at the base of the melt region and crystallization proceeds from the bottom up. When the temperature decreases to the rheological transition temperature, a partially solid region develops (light gray) and with further cooling, a rheological front propagates through the melt region (dark gray). (Bottom) Crystallization timescale for surface recycling and stagnant lid convection as a function of melt region size. The shaded region corresponds to a viscosity uncertainty factor of ±10 and is indicated on timescale graphs throughout the paper.

rheological variations within the flow and plastic yielding in the crust at the flow periphery can control the lateral spreading timescale.

### 3.1. *What is the timescale of local magma ocean crystallization?*

Vigorous convection in the low viscosity melt region is the primary mode of cooling and crystallization. As the region cools and crystallizes, a partially solid layer (corresponding to the rheological transition, Section A.3) develops from the bottom up (Fig. 3). The cooling efficiency depends on stability of a stagnant lid which can form near the surface of the convecting magma (Davaille and Jaupart, 1993).

#### 3.1.1. *Convection with surface recycling in the local magma ocean*

One possible scenario for crystallization is that the cold boundary layer at the surface of the melt region is able to subduct and participate in the convective flow. For low viscosity liquid silicates, convection is in a turbulent regime (Kraichnan, 1962; Shraiman and Siggia, 1990; Siggia, 1994). Thermal evolution of the melt region depends on the mean global heat flux, $F$, which is a function of the Rayleigh number and Prandtl number. The Rayleigh and Prandtl numbers are measures of the vigor of convection and the ratio of momentum to thermal diffusivity, respectively. The effect of finite Prandtl number on turbulent plume development has recently been addressed by Hier-Majumder et al. (2004). For the magma ocean, the dynamic viscosity $\nu_l = \eta_l/\rho_l \sim 3\times10^{-5}$ m$^2$/s and thermal diffusivity $\kappa \sim 10^{-6}$ m$^2$/s (Table 1) implying a Prandtl number $\mathrm{Pr} = \nu_l/\kappa \sim 30$. For turbulent convection and $\mathrm{Pr} > 1$, mixing length theory (Kraichnan, 1962; Siggia, 1994) gives a Pr independent heat flux,

$$F = 0.089k(T - T_s)^{4/3}\left(\frac{\rho_l \alpha g}{\eta_l \kappa}\right)^{1/3}, \tag{13}$$

where $k$ is the thermal conductivity, $\rho_l$ is the silicate liquid density, $\alpha$ is the coefficient of thermal expansion, $\eta_l$ is the silicate liquid viscosity, $\kappa$ is the thermal diffusivity, and $T$ is the potential temperature. The convective heat flux must match the surface heat flux, which, in a radiative blackbody approximation, is

$$F = \sigma_B T_s^4, \tag{14}$$

where $\sigma_B$ is the Stefan–Boltzmann constant and $T_s$ is the surface temperature. At a potential temperature of 2000 K, corresponding to superliquidus conditions, the surface temperature and heat flux are ∼1400 K and $3\times10^5$ W/m$^2$, respectively. The timescale for crystallization to the critical melt fraction is

$$t_{\mathrm{crys}} \approx \frac{[L(1-\phi_{cr}) + c_p \Delta T_{cr}]\rho_l V}{F\pi R^2}, \tag{15}$$

where $L$ is the latent heat of melting, $c_p$ is the specific heat, and the surface area of the melt region $\pi R^2$. For the variation of melt fraction between liquidus and solidus suggested by the experimental data of McKenzie and Bickle (1988), the temperature change corresponding to $\Delta\phi = 1 - \phi_{cr}$ is $\Delta T_{cr} \approx 0.4(T_l - T_s) \approx 0.4(2000 - 1400)$ K $\approx 240$ K. The crystallization time for $\xi \sim 0.5$ is $t \sim 300$ yr (Fig. 3, Table 2). If convection is in the hard turbulence regime (Shraiman and Siggia, 1990; Siggia, 1994), the heat flux is slightly higher and weakly dependent on both magma ocean depth and Pr. However, the crystallization timescale estimated in this section does not change significantly. The insulating effect of an impact induced steam atmosphere reduces the heat flux by 2–3 orders of magnitude (Matsui and Abe, 1986; Zahnle et al., 1988) which would increase the crystallization time by the same factor.

#### 3.1.2. *Convection beneath a stagnant lid in the local magma ocean*

The large temperature contrast across the upper thermal boundary layer of the convecting magma region implies a large viscosity variation and the possible development of a stable, stagnant lid (Davaille and Jaupart, 1993). In this case, the convective driving temperature is reduced. If the crystallizing magma behaves like a slurry, the rheology can be described in terms of an effective viscosity which depends on temperature and crystal fraction. Variation of viscosity near the liquidus is used as a first approximation, and it follows that the Einstein model for the Newtonian viscosity of dilute suspensions (Einstein, 1906) can be used,

$$\eta_l(\phi) \sim \exp\left(\frac{E_m}{RT}\right)\left(1 + \frac{5}{2}(1-\phi)\right), \tag{16}$$

where $E_m$ is the activation energy and $R$ is the gas constant. The rheological temperature scale which drives convection beneath the lid

Table 1
Notation

| Parameter | Symbol | Value | Dimensions |
|---|---|---|---|
| Impactor radius | $a$ | | km |
| Critical impactor radius | $a_{cr}$ | 700–1500 | km |
| Non-Newtonian coefficient (partial melt) | $b_m$ | $10^{30}$ | $\mathrm{Pa}^3\,\mathrm{s}$ |
| Non-Newtonian coefficient (solid) | $b_s$ | $10^{37}$ | $\mathrm{Pa}^3\,\mathrm{s}$ |
| Cratering parameter | $C_D$ | 1.6 | None |
| Linear shock parameter | $c$ | 4.4 | $\mathrm{km\,s^{-1}}$ |
| Thermal capacity | $c_p$ | 1200 | $\mathrm{J\,kg^{-1}\,K^{-1}}$ |
| Modified Stokes drag force | $D$ | | N |
| Stokes drag force | $D_n$ | | N |
| Grain size | $D_p$ | 1 | mm |
| Melt activation energy | $E_m$ | 250 | $\mathrm{kJ\,mole^{-1}}$ |
| Convective heat flux | $F$ | | $\mathrm{W\,m^{-2}}$ |
| Modified Stokes buoyancy force | $F_b$ | | N |
| Retained melt fraction | $f$ | | None |
| Critical retained melt fraction | $f_{cr}$ | 0.5–0.6 | None |
| Modified Stokes volume fraction | $f_V$ | | None |
| Modified Stokes area fraction | $f_S$ | | None |
| Gravitational constant | $G$ | $6.67 \times 10^{-11}$ | $\mathrm{N\,m^2\,kg^{-2}}$ |
| Gravitational acceleration | $g$ | 3.7 | $\mathrm{m\,s^{-2}}$ |
| Thermal conductivity | $k$ | 4 | $\mathrm{W\,m^{-1}\,K^{-1}}$ |
| Specific permeability | $k_\phi$ | | $\mathrm{m^2}$ |
| Silicate latent heat of fusion | $L$ | 500 | $\mathrm{kJ\,kg^{-1}}$ |
| Mars mass | $M$ | $6.4 \times 10^{23}$ | kg |
| Impact shock pressure | $P$ | | GPa |
| Gravity current volume flux ($= V/t_{\mathrm{iso}}$) | $Q$ | | $\mathrm{m^3\,s^{-1}}$ |
| Partial melt region radius | $R$ | | km |
| Mars radius | $R_p$ | 3398 | km |
| Gas constant | $R^*$ | 8.314 | $\mathrm{J\,mole^{-1}\,K^{-1}}$ |
| Transient crater radius | $r_c$ | | km |
| Complete melt radius | $r_m$ | | km |
| Gravity current radius | $S$ | | km |
| Entropy | S | | $\mathrm{J\,kg^{-1}\,K^{-1}}$ |
| Linear shock parameter | $s$ | 1.5 | None |
| Temperature | $T$ | | K |
| Surface temperature | $T_S$ | 220 | K |
| Solidus temperature | $T_s$ | | K |
| Liquidus temperature | $T_l$ | | K |
| Temperature change between $\phi = \phi_{cr}$ and $\phi = 0$ | $\Delta T$ | 400 | K |
| Temperature change between $\phi = 1$ and $\phi_{cr}$ | $\Delta T_{cr}$ | 240 | K |
| Rheological temperature scale, partial melt | $\Delta T_d$ | 100–200 | K |
| Rheological temperature scale, complete melt | $\Delta T_{d,m}$ | 100 | K |
| Time | $t$ | | s |
| Gravity current velocity | $U$ | | $\mathrm{km\,yr^{-1}}$ |
| Partial melt region volume | $V$ | | $\mathrm{km^3}$ |
| Transient crater volume | $V_c$ | | $\mathrm{km^3}$ |
| Complete melt region volume | $V_m$ | | $\mathrm{km^3}$ |
| Excavated melt volume | $V_{m,e}$ | | $\mathrm{km^3}$ |
| Mars volume | $V_p$ | $1.6 \times 10^{11}$ | $\mathrm{km^3}$ |
| Percolation velocity | $v$ | | $\mathrm{m\,s^{-1}}$ |
| Impact velocity | $v_i$ | 7 | $\mathrm{km\,s^{-1}}$ |
| Impact velocity for shock pressure $P$ | $v_i^P$ | | $\mathrm{km\,s^{-1}}$ |
| Gravity current height | $Z$ | | km |
| Transient crater exponent parameter | $z$ | 3 | None |
| Thermal expansion | $\alpha$ | $2 \times 10^{-5}$ | $\mathrm{K^{-1}}$ |
| Cratering parameter | $\beta$ | 0.22 | None |
| Crater depth/radius ratio | $\gamma_c$ | 1/2–2/3 | None |
| Complete melt region-impactor radius ratio ($= r_m/a$) | $\gamma_m$ | 1.3 | None |
| Partial melt region-impactor radius ratio ($= R/a$) | $\gamma$ | 1.7 | None |
| Gravity current crustal thickness | $\delta$ | | km |
| Compaction lengthscale | $\delta_c$ | | km |
| Crustal viscosity | $\eta_c$ | | Pa s |

(*continued on next page*)

Table 1 (*continued*)

| Parameter | Symbol | Value | Dimensions |
|---|---|---|---|
| Newtonian viscosity (solid) | $\eta_s$ | $10^{21}$ | Pa s |
| Newtonian viscosity (partial melt) | $\eta_m$ | $10^{17}$ | Pa s |
| Liquid silicate viscosity | $\eta_l$ | 0.1 | Pa s |
| Melt viscosity (near solidus) | $\eta_f$ | 100 | Pa s |
| Thermal diffusivity | $\kappa$ | $10^{-6}$ | $m^2\,s^{-1}$ |
| Geometrical factor ($= 16/13(1-3/8\xi)$) | $\Xi$ | | None |
| Partial melt region size ($= R/R_p$) | $\xi$ | | None |
| Complete melt region size ($= r_m/R_p$) | $\xi_m$ | | None |
| Solid silicate density | $\rho$ | 3500 | $kg\,m^{-3}$ |
| Silicate density ($@\phi_{cr}$) | $\rho_{cr}$ | 3400 | $kg\,m^{-3}$ |
| Liquid silicate density | $\rho_l$ | 3200 | $kg\,m^{-3}$ |
| Stefan–Boltzmann constant | $\sigma_B$ | $5.67 \times 10^{-8}$ | $W\,m^{-2}\,K^{-4}$ |
| Yield stress | $\tau_y$ | $10^7$ | Pa |
| Isostatic adjustment-crystallization time ratio | $\Upsilon_{\text{iso/crys}}$ | | None |
| Lateral spreading-conductive cooling time ratio | $\Upsilon_{\text{sprd/cool}}$ | | None |
| Melt fraction | $\phi$ | | None |
| Critical melt fraction | $\phi_{cr}$ | 0.4 | None |

Table 2
Timescales

| Process | Equation |
|---|---|
| Crystallization (surface recycling) | $3 \times 10^2\ \text{yr}(\eta_l/0.1\ \text{Pa s})^{1/3}(\xi/0.5)\Xi$ |
| Crystallization (stagnant lid) | $10^3\ \text{yr}(\eta_l/0.1\ \text{Pa s})^{1/3}(\xi/0.5)\Xi$ |
| Convective cooling (stagnant lid) | $3 \times 10^8\ \text{yr}(F/100\ \text{mW/m}^2)^{-1}(\xi/0.5)\Xi$ |
| Convective cooling (surface recycling) | $10^8\ \text{yr}(F/1\ \text{W/m}^2)^{-1}(\xi/0.5)^{1.06}\Xi$ |
| Compaction | $1\ \text{yr}(\eta_m/10^{17}\ \text{Pa s})(\xi/0.5)^{-1}$ |
| Differentiation | $10^8\ \text{yr}(0.02/\phi)^2(\eta_f/100\ \text{Pa s})(\xi/0.5)$ |

*Note*. Geometrical factor $\Xi = 16/13(1-3/8\xi)$, $\Xi = 1$ at $\xi = 0.5$.

$$\Delta T_{d,m} \sim \left|\frac{\partial \ln \eta}{\partial T}\right|^{-1} \sim \left[\frac{E_m}{RT^2} + \left(1 + \frac{5}{2}(1-\phi)\right)^{-1}\frac{5}{2}\left|\frac{\partial \phi}{\partial T}\right|\right]^{-1}. \tag{17}$$

At the liquidus, $\phi = 1$ and to first order at zero pressure, $|\partial\phi/\partial T| \sim 1/500$ K. The activation energy of olivine tholeiite melt at 1 GPa $E_m \sim 250$ kJ/mol (Kushiro, 1986). Thus, the driving temperature for convection is ∼100 K which is an order of magnitude smaller than the surface recycling case. The convective heat flux in the stagnant lid regime (Solomatov and Moresi, 2000),

$$F = 0.53k\Delta T_{d,m}^{4/3}\left(\frac{\rho_l \alpha g}{\eta_l \kappa}\right)^{1/3}. \tag{18}$$

For $\Delta T_{d,m} \sim 100$ K, $F \sim 10^5$ W/m$^2$, and the crystallization time for $\xi \sim 0.5$ is $t \sim 10^3$ yr (Fig. 3, Table 2).

### 3.2. *What is the timescale for isostatic adjustment to a local magma ocean?*

Because the melt region density is less than the density of solid silicates, the melt region is positively buoyant and the planet deforms in order to restore isostatic equilibrium. Isostatic adjustment involves both a radial and lateral component, i.e., the melt region moves upward and spreads laterally until spherical symmetry is restored. The timescale for radial relaxation is controlled by the rheology of the solid mantle and is estimated from a modified Stokes' flow model. The lateral component of isostatic adjustment involves deformation of the melt region itself and the timescale is estimated from an axisymmetric, plastic crust controlled, gravity current.

#### 3.2.1. *Radial relaxation of the planet in response to the local magma ocean*

Melt region buoyancy results in viscous deformation of the planet and radial motion of the melt toward the surface in order to restore an isostatic mass distribution. An order of magnitude estimate of the radial relaxation timescale is derived from a modified Stokes' flow model similar to one developed for core formation by Honda et al. (1993). The buoyancy and drag forces on the melt region of radius $R$ are

$$F_b = \frac{4}{3}\pi R^3(\rho - \rho_l)g f_V(r) \tag{19}$$

and

$$D = D_n f_S(r), \tag{20}$$

respectively. Here, $f_V(r)$ and $f_S(r)$ are the volume and surface area fractions of a sphere of radius $R$ embedded in a sphere of radius $R_p$ when the sphere centers are separated by a distance $r$ (Fig. 4). $D_n$ is the drag force on an inviscid sphere of radius $R$ moving with a velocity $v_{\text{iso}}$,

$$D_n = \begin{cases} 4\pi R\eta_s v_{\text{iso}}, & n = 1, \\ 8.6\pi R^2(b_s v_{\text{iso}}/R)^{1/3}, & n = 3. \end{cases} \tag{21}$$

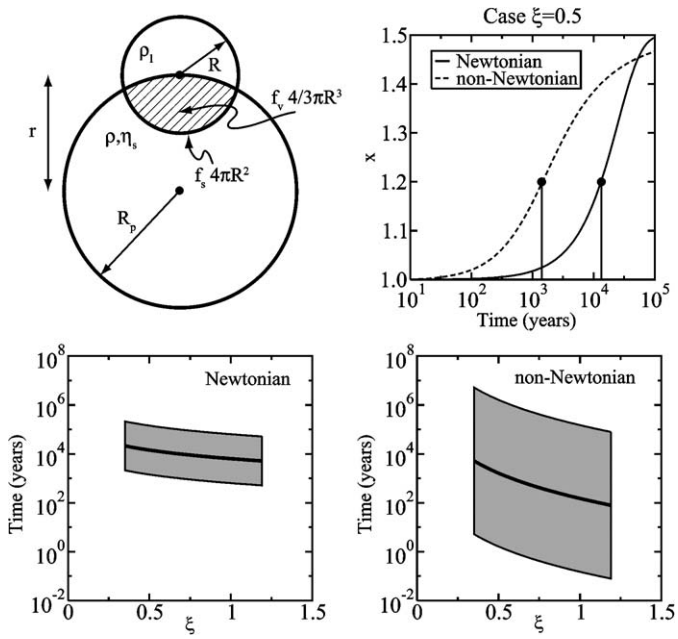


Fig. 4. (Top, left) Radial relaxation model. The cross-hatched region corresponds to the melt region and the volume and surface area fractions are $f_V$ and $f_S$, respectively. The driving density difference is $\rho - \rho_l$ and the relaxation velocity $v_{\text{iso}} = R_p\dot{x}$, where $x = r/R_p$. Relaxation is controlled by the viscosity of the solid part of the planet, $\eta_s$. (Top, right) Results for $\xi = 0.5$. Location of the melt region $r$ versus time for Newtonian (solid line) and power law viscosity (dashed line) corresponding to solid mantle. Complete adjustment corresponds to $x = 1+\xi = 1.5$. The adjustment timescale is defined as the time when the embedded melt region volume drops by a factor of $e$ from the initial value. For $\xi = 0.5$, this corresponds to $x \sim 1.2$ (solid circles) and the times are indicated with vertical lines. (Bottom) Isostatic relaxation time as a function of melt region size for Newtonian (left) and power law (right) viscosity.

The power law viscosity ($n = 3$) result is from the variational principle approach of Lliboutry (1987). Adopting some notation of Honda et al. (1993), the volume and area fractions are given by

$$f_V = \frac{3}{4}\left[c_1^2\left(1 - \frac{c_1}{3}\right) + c_2^2\left(\frac{1}{\xi} - \frac{c_2}{3}\right)\right] \tag{22}$$

and

$$f_S = \frac{c_1}{2}, \tag{23}$$

where

$$c_1 = \frac{1 - (\xi - x)^2}{2\xi x}, \tag{24}$$

$$c_2 = \frac{\xi^2 - (1 - x)^2}{2\xi x}, \tag{25}$$

with $x = r/R_p$. Balancing the buoyancy and drag force yields the equation of motion for the buoyant melt region. For Newtonian viscosity

$$\dot{x} = \frac{1}{3}\frac{R_p(\rho - \rho_l)g}{\eta}\xi^2\frac{f_V}{f_S}, \tag{26}$$

while for power law viscosity

$$\dot{x} = 3.7 \times 10^{-3}\frac{(R_p(\rho - \rho_l)g)^3}{b}\xi^4\left(\frac{f_V}{f_S}\right)^3. \tag{27}$$

Note that the factor $f_V/f_S$ in these equations implies that isostatic adjustment velocity $v_{\text{iso}} = R_p\dot{x}$ is not constant but varies as relaxation proceeds. As $x \to 1 + \xi$, $f_S$ and $f_V$ go to zero linearly and quadratically, respectively so the driving force decreases faster than the drag force and late stage relaxation can take a long time. The radial relaxation time, $t_{\text{iso}}$, is approximated by the time for the embedded volume to drop by a factor of $e$, i.e., the time when

$$2f_V e = (1 - 3/8\xi). \tag{28}$$

This time should be considered a lower bound because late stage relaxation can contribute (Fig. 4) and because $\xi$ corresponds to the region where $\phi$ varies from 1 to $\phi_{cr}$ so the effective density difference is smaller than the solid – pure liquid difference. The radial relaxation timescale for $\xi \sim 0.5$ is $\sim 10^4$ yr for both Newtonian and non-Newtonian viscosity. For larger $\xi$, the timescale can drop by a factor of $\sim 100$ in the non-Newtonian case (Fig. 4).

For the largest magma ocean sizes, plastic deformation may control the radial relaxation. For a fully molten blob on one side of Mars with $\xi \sim 1$–1.2, the differential stress $(\rho - \rho_l)gR_p \sim$ 3 GPa. This exceeds the power law creep breakdown stress (Tsenn and Carter, 1987) of $\sim 10^{-2}\mu$, where $\mu$ is the shear modulus. Above this transition stress, the strain rate increases rapidly with increasing stress (best described with an exponential stress–strain rate relation). In this case, the radial relaxation time would be significantly reduced. Furthermore, this differential stress is close to the ultimate strength of rocks which is $\sim$1–2 GPa (Davies, 1982; Kinsland, 1978). If the ultimate strength is exceeded, radial relaxation could be an extremely fast process.

### 3.2.2. *Lateral melt flow at the surface*

Melt extruded onto the surface during isostatic adjustment spreads laterally around the planet. Here, surface spreading is considered in the context of an axisymmetric, constant flux, divergent, gravity current model (Huppert, 1982; Gratton et al., 1999). Volume flux is estimated from the local magma ocean volume and radial relaxation timescale. Exposure of high temperature liquid silicates to the ambient atmospheric temperature results in a large, radiative surface heat flux and rapid quenching of a thin, solid surface layer. This has important implications for spreading dynamics. A flow model with rheologically uniform interior and thin, solidified, crustal surface layer is adopted (Griffiths and Fink, 1993). The viscosity contrast between the interior and surface is large and crustal stresses can exceed the material yield strength. One simple approach to the spreading problem with material yielding is to treat the material as viscoplastic such that crustal stress saturates at a critical yield stress, $\tau_y$. In this case, spreading is in a plastic crust controlled regime.

For an axisymmetric, constant volume flux, plastic crust controlled, gravity current (Griffiths and Fink, 1993), the flow radius

$$S \sim \left(\frac{\rho_l g Q^2}{\tau_y \kappa^{1/2}}\right)^{1/4} t^{3/8}, \tag{29}$$

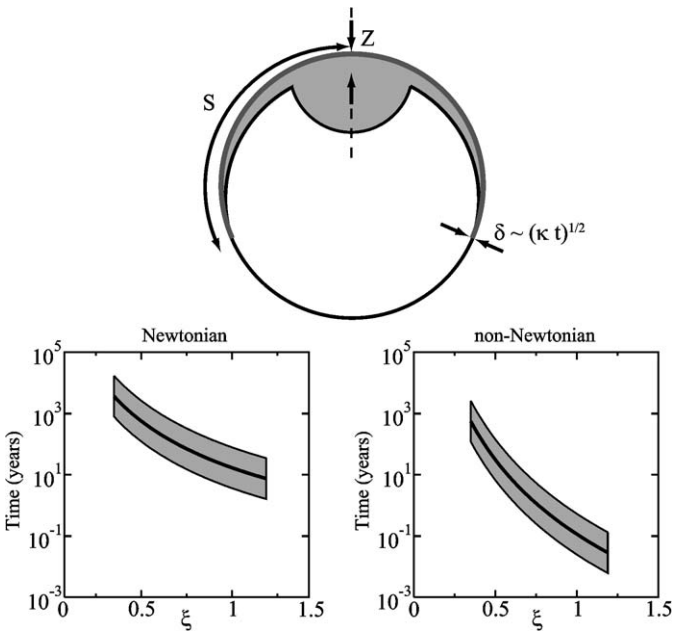


Fig. 5. (Top) Lateral spreading model. The dark shaded region corresponds to melt and the lighter shaded region to the cold surface boundary layer of solidified crust with thickness $\delta \sim (\kappa t)^{1/2}$. The central height and radius of the gravity current are $Z$ and $S$, respectively. (Bottom) Lateral spreading time as a function of melt region size for Newtonian (left) and power law (right) radial relaxation models.

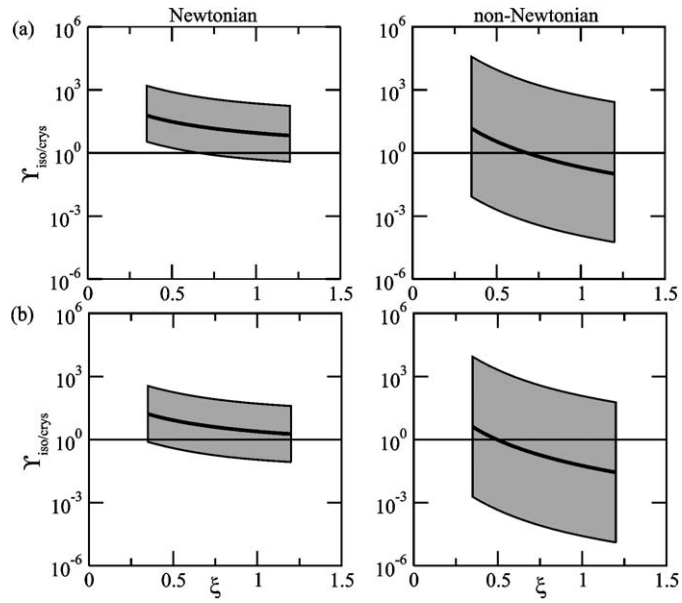


Fig. 6. The ratio of the isostatic adjustment time to the crystallization time $\Upsilon_{\mathrm{iso/crys}}$ for Newtonian and non-Newtonian solid mantle rheology and convection with (a) surface recycling and (b) a stable stagnant lid.

where $Q$ is the volume flux. An important parameter is the yield strength $\tau_y$. The low pressure yield strength of rocks (cohesive strength) is ∼10–100 MPa and increases with pressure (frictional law). For spreading around a spherical planet, the gravity current makes a transition from a divergent regime to a convergent regime when the flow covers one hemisphere. While the details of the flow are different in the convergent case (Diez et al., 1992), the spreading time of the diverging flow and so-called closure time for the converging flow are of the same order. Thus, the timescale for lateral spreading is estimated from the criterion that the flow area equal the planetary surface area $S^2 \sim 2R_p^2$. For $\xi \sim 0.5$ the melt volume $V \sim 0.05V_p$ [Eq. (5)] and the radial relaxation time is $\sim 10^4$ yr (Fig. 4). For this magma ocean size, $Q \sim 3 \times 10^7$ m$^3$/s and the lateral spreading timescale $t_{\mathrm{sprd}} \sim 10^3$ yr (Fig. 5) a factor of ∼10 smaller than the radial relaxation time. This is the case for all $\xi$ and thus isostatic adjustment is limited by radial relaxation. In the next section, this time is compared to the in situ crystallization time.

The validity of the scaling can be estimated by comparing the magnitude of various forces to the crustal yield strength. Crustal yielding requires, $\eta_c U/Z \gg \tau_y$, where $\eta_c$ is the crustal viscosity and $U \sim S/t$ is the flow velocity. Given the scaling [Eq. (29)], the criterion holds throughout spreading provided $\eta_c \gg (\kappa^2 \tau_y^7 R_p^7/(Q^5 \rho_l^4 g^4))^{1/3}$ which for the smallest $Q$ implies $\eta_c \gg 10^{12}$ Pa s. This is probably well satisfied because the effective crustal viscosity is heavily weighted toward the viscosity of the cold boundary layer (Stasuik et al., 1993; Griffiths and Fink, 1993; Lister and Kerr, 1993). For the crustal yield strength to dominate the basal viscous stress throughout spreading, $\eta_l \ll (\kappa^4 \tau_y^8 R_p^8/(Q^7 \rho_l^5 g^5))^{1/3}$. For the nominal value $\eta_l \sim 0.1$ Pa s, this condition is met for all but the largest $\xi$ in the non-Newtonian radial relaxation model.

### 3.3. Comparison of isostatic adjustment time and crystallization time

Formation of a global magma ocean depends on the ratio of isostatic adjustment time to crystallization time

$$\Upsilon_{\mathrm{iso/crys}} \sim \frac{t_{\mathrm{iso}}}{t_{\mathrm{crys}}}. \tag{30}$$

For $\Upsilon_{\mathrm{iso/crys}} \ll 1$, melt is extruded and spreads around the planet before in situ crystallization. For $\Upsilon_{\mathrm{iso/crys}} \gg 1$, the melt region crystallizes to a partially molten state before redistribution can occur.

The value of $\Upsilon_{\mathrm{iso/crys}}$ depends on mode of crystallization, solid mantle rheology, and melt region size (Fig. 6). Since $t_{\mathrm{crys}}$ increases faster than $t_{\mathrm{iso}}$ with increasing $\xi$, large melt regions ($\xi \sim 1$), tend to have $\Upsilon_{\mathrm{iso/crys}} < 1$ independent of crystallization mechanism. The exception is Newtonian viscosity and efficient convective cooling with surface recycling. Within rheological uncertainties, small melt regions tend to have $\Upsilon_{\mathrm{iso/crys}} > 1$. In this case, the melt region crystallizes to the critical crystal fraction before the planet adjusts isostatically. The subsequent evolution of such a region is discussed in Section 4. The critical parameter controlling whether the initial melt region becomes a global magma ocean or local partial melt pond is the rheology of the solid part of the planet. The effective viscosity of the mantle is strongly temperature dependent (Section A.1) and thus Mars thermal state at the time of the impact is expected to play a crucial role.

For the largest magma ocean sizes considered, $\xi \sim 1$–1.2, the differential stress driving isostatic adjustment approaches

the transition stress for power-law creep breakdown (Tsenn and Carter, 1987) and the ultimate strength of rocks (Davies, 1982). In this case, the viscous model considered here breaks down but the main conclusion remain unchanged. For plastic deformation, isostatic adjustment would be a much faster process than crystallization, i.e., $\Upsilon_{\mathrm{iso/crys}} < 1$. This is predicted by the viscous model for $\xi > 1$ (Fig. 6).

## 4. Evolution of a local magma ocean after initial crystallization to a partially molten state

In this section, evolution of the local magma ocean subsequent to in situ crystallization is considered. After the magma ocean crystallizes to the critical crystal fraction, several changes occur which affect the evolution. One factor is that the driving density difference for radial relaxation is reduced. A more significant effect is that the rheology of partially molten material (at or above the critical crystal fraction) is close to that of solid silicates. Deformation of partially molten silicates by thermally activated creep is characterized by an effective viscosity which is smaller than that of solids (Section A.2) but ∼20 orders of magnitude larger than silicate liquid viscosity. It follows that the convective cooling timescale increases dramatically. In addition to a solid-like viscosity, the partially molten material may fail at sufficiently high stresses. Stresses associated with lateral spreading of partially molten material can exceed the material yield strength. This is the case above a critical volume flux (corresponding to a critical local magma ocean size) and lateral spreading is controlled by plastic deformation at the base of the flow. Below the critical volume flux, spreading is controlled by viscous stresses.

### *4.1. Solid-state convective cooling in the partially molten local magma ocean*

Convection in the partially molten solid is the primary mechanism of cooling and crystallization below the critical melt fraction. The timescale to cool the region with $0 < \phi < \phi_{cr}$ to the solidus is

$$t_{\mathrm{conv}} \approx \frac{[L\phi_{cr} + c_p \Delta T]\rho_{cr} V}{F \pi R^2}, \tag{31}$$

$\rho_{cr}$ is the density at the critical melt fraction and $F$ is the convective heat flux. The temperature drop corresponding to $\Delta\phi = \phi_{cr}$ is $\Delta T \approx 0.6(T_l - T_s) \approx 400$ K. As with initial crystallization, the heat flux $F$ depends on the efficiency of surface recycling.

In the stagnant lid regime, the heat flux for general $n$ (Solomatov and Moresi, 2000),

$$F \approx (0.31 + 0.22n) k \Delta T_d \left( \frac{\rho_{cr} g \alpha \Delta T_d}{b_m^{1/n} \kappa^{1/n}} \right)^{n/(n+2)}, \tag{32}$$

where $n$ is the power law exponent, $\Delta T_d$ is the driving temperature scale,

$$\Delta T_d = \left| \frac{\partial \ln \eta}{\partial T} \right|^{-1} \sim 100\text{–}200 \text{ K}, \tag{33}$$

and $b_m$ is the power law viscosity coefficient ($b_m = \eta_m$ when $n = 1$) appropriate for partially molten mantle. For $\Delta T_d \sim$ 100 K and nominal, partially molten rheological parameters, $F \sim 100$ and 30 mW/m$^2$ for $n = 1$ and 3, respectively. For $\xi \sim 0.5$ and Newtonian viscosity, the cooling time is ∼300 Myr (Fig. 7, Table 2).

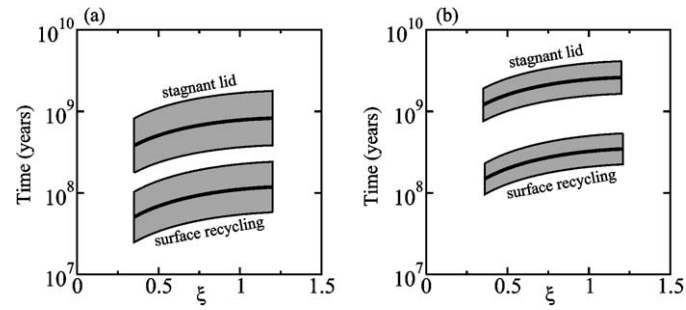


Fig. 7. Timescale for convective cooling of the partial melt region beyond the critical crystal fraction for (a) Newtonian and (b) non-Newtonian rheology as a function of melt region size. If surface recycling occurs, the driving temperature for convection $\Delta T_d \sim 1000$ K and the cooling timescale is ∼100–300 Myr. If surface recycling is inefficient, convection occurs in the stagnant lid regime and the driving temperature $\Delta T_d \sim 100$ K. The cooling timescale is ∼300 Myr–2 Gyr.

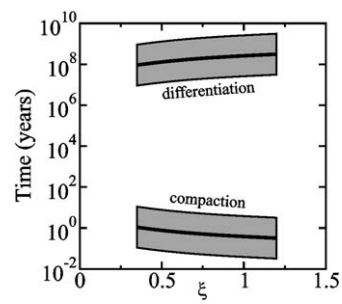


Fig. 8. Time for percolation to reduce the melt fraction from $\phi_{cr}$ to ∼2% and compaction timescale as a function of melt region size.

In the surface recycling (small viscosity contrast) regime, the heat flux for general $n$ (Solomatov, 1995; Reese et al., 1998),

$$F = 0.26 \frac{k \Delta T_d}{d} \left( \frac{\rho_{cr} \alpha g \Delta T_d d^{(n+2)/n}}{b_m^{1/n} \kappa^{1/n}} \right)^{0.94n/(n+2)}, \tag{34}$$

where $d$ is the layer depth. When the entire upper cold boundary layer participates in the convective flow, the driving temperature is $\Delta T_d \sim 1000$ K and $F \sim 1$ and 300 mW/m$^2$ for $n = 1$ and 3, respectively. For $\xi \sim 0.5$ and Newtonian viscosity, the cooling time is ∼100 Myr (Fig. 7, Table 2).

### *4.2. Melt migration in the partially molten local magma ocean*

#### *4.2.1. Compaction of partially molten material at the bottom of the local magma ocean*

Length and timescales for compaction (McKenzie, 1984; Bercovici et al., 2001) in the partially molten magma ocean are estimated from a simple model. In the two-phase region with solid density $\rho$ and melt density $\rho_l$, the driving pressure for self-compaction for a system depth $R$ is $(\rho - \rho_l) g R$. This pressure is balanced by the shear stress associated with deformation of the solid matrix, $\tau \sim \dot{\phi}/\phi$. The compaction timescale is esti-

mated as the time for $\phi$ to drop by a factor of $e$,

$$t_{\text{comp}} \sim \frac{\eta_m}{(\rho - \rho_l) g R}. \tag{35}$$

For $\xi \sim 0.5$, the compaction timescale is $\sim$10 yr (Fig. 8, Table 2). Compaction occurs over a length scale $\delta_c$ at the relatively impermeable bottom of the local magma ocean. Darcy's law gives a melt velocity $k_\phi/\eta_f(\rho - \rho_l)gR/\delta_c$. Thus, the compaction length

$$\delta_c \sim \left(k_\phi \frac{\eta_m}{\eta_f}\right)^{1/2}, \tag{36}$$

where $k_\phi$ is the specific permeability and $\eta_f$ is the near solidus melt viscosity. In the low Reynolds number limit, the Ergun equation for $k_\phi$ (Ergun, 1952) reduces to the Blake–Carman–Kozeny equation (e.g., Bird et al., 1960),

$$k_\phi = \frac{\phi^3 D_p^2}{150(1-\phi)^2}, \tag{37}$$

where $D_p$ is the grain size. For $D_p \sim 1$ mm and $\phi = \phi_{cr}$, $k_\phi \sim 10^{-9}$ m$^2$ and $\delta_c \sim 1$ km which is small for the system sizes under consideration. Thus, melt migration dynamics are determined mainly by percolation.

#### 4.2.2. *Melt percolation and differentiation in the partially molten local magma ocean*

Above the compaction region, gravitational body forces are balanced by viscous stresses. The percolation velocity of melt through the solid matrix

$$v_{\text{perc}} = \frac{k_\phi(\rho - \rho_l)g}{\eta_f}\frac{(1-\phi)}{\phi} = \frac{(\rho - \rho_l)gD_p^2}{150\eta_f}\frac{\phi^2}{(1-\phi)}, \tag{38}$$

where the second equality follow from Eq. (37). The differentiation time

$$t_{\text{diff}} \sim \frac{R}{v_{\text{perc}}} \sim \frac{150\eta_f R}{(\rho - \rho_l)gD_p^2}\frac{1-\phi}{\phi^2}. \tag{39}$$

For $\xi \sim 0.5$, the time to reduce the melt fraction to 2% is $\sim 10^8$ yr (Fig. 8, Table 2).

### 4.3. *What is the timescale for isostatic adjustment to a partially molten local magma ocean?*

#### 4.3.1. *Radial relaxation of the planet in response to the partially molten local magma ocean*

Crystallization reduces the density difference which drives radial relaxation. At the critical melt fraction, $\phi_{cr}$, the driving density difference $\rho - \rho_{cr} = (\rho - \rho_l)\phi_{cr} \sim 100$ kg/m$^3$. Thus, the radial relaxation model is applicable with timescales increased by a factor of $(\rho - \rho_l)/(\rho - \rho_{cr}) \sim \phi_{cr}^{-1} \sim 3$ (Fig. 9).

#### 4.3.2. *Lateral spreading of partially molten silicates at the surface*

Partially molten material extruded onto the surface spreads laterally to complete the process of isostatic adjustment. This material has a rheological behavior close to solid silicates because it has crystallized to the critical melt fraction. For nominal parameters (Table 1), basal stress can exceed the yield stress.

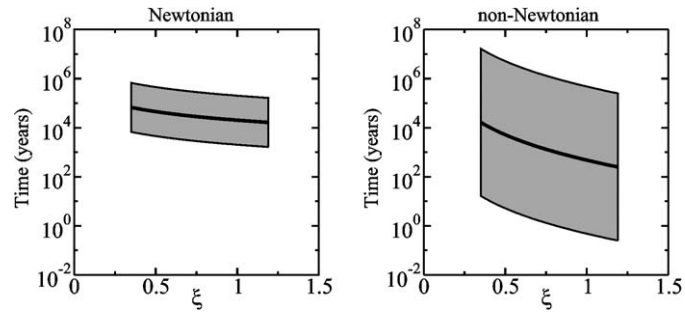


Fig. 9. Radial relaxation time for Newtonian (left) and power-law (right) viscosity subsequent to initial crystallization.

For a plastic axisymmetric gravity current (Nye, 1951; Blake, 1990), the radius

$$S \sim \left(\frac{\rho_{cr} g Q^2}{\tau_y}\right)^{1/5} t^{2/5}. \tag{40}$$

The spreading time for Newtonian radial relaxation, $\tau_y \sim 10$ MPa and $\xi \sim 0.75$ is $\sim 10^4$ yr which is approximately of the same order as the radial relaxation time (Table 2, Fig. 10). This is approximately the case for all $\xi$ in this regime implying that lateral spreading and radial relaxation occur essentially simultaneously during isostatic adjustment. The thickness of the flow $Z \sim (\tau_y Q^{1/2}/\rho_{cr} g)^{2/3} t^{1/5}$. Thus, at the end of isostatic adjustment, for Newtonian radial relaxation and $\xi \sim 0.75$, the central height $Z \sim 100$ km and the thickness decreases with distance from the flow axis. This asymmetric mass distribution would continue to relax viscously with time.

There is a transition in spreading regime for Newtonian radial relaxation. For the basal viscous stress to exceed the basal yield stress throughout spreading given the nominal value of $\eta_m$ (Table 1), the volume flux $Q \gg 4\pi^{2/5} R_p^2 \tau_y^2/(\rho_{cr} g \eta_m) \sim 6 \times 10^6$ m$^3$/s. This condition is met for all $\xi$ and non-Newtonian radial relaxation but a transition occurs for Newtonian radial relaxation. For $\xi < 0.5$, the volume flux drops below the critical one for basal yielding and spreading is in a viscous regime. Another important implication of this transition is that the spreading timescale becomes larger than the radial relaxation time and spreading is more appropriately modeled as a constant volume flow rather than a constant volume flux flow. For an axisymmetric, viscous, constant volume, divergent gravity current the radius

$$S \sim \pi^{-3/8}\left(\frac{\rho_{cr} g V^3}{\eta_m}\right)^{1/8} t^{1/8}. \tag{41}$$

The spreading timescale in this regime increases from about $10^5$ yr near the transition to $10^7$ yr at the low end of the $\xi$ range [Fig. 10(a)]. The bulk viscosity of the flow increases with time due to cooling which slows and eventually stops the flow (Sakimoto and Zuber, 1995; Bercovici and Lin, 1996). Spreading should essentially stop after approximately one diffusion time. For cooling from above and below this time satisfies $Z/2 \sim 2(\kappa t)^{1/2}$, where $Z$ is the flow thickness. In the constant volume, viscous regime $Z$ decreases with time, $Z \sim (\eta_m V/(\rho_{cr} g))^{1/4} t^{-1/4}$. The cooling time in this case is $\sim 10^6$ yr [Fig. 10(b)]. The competing effects of lateral spreading and

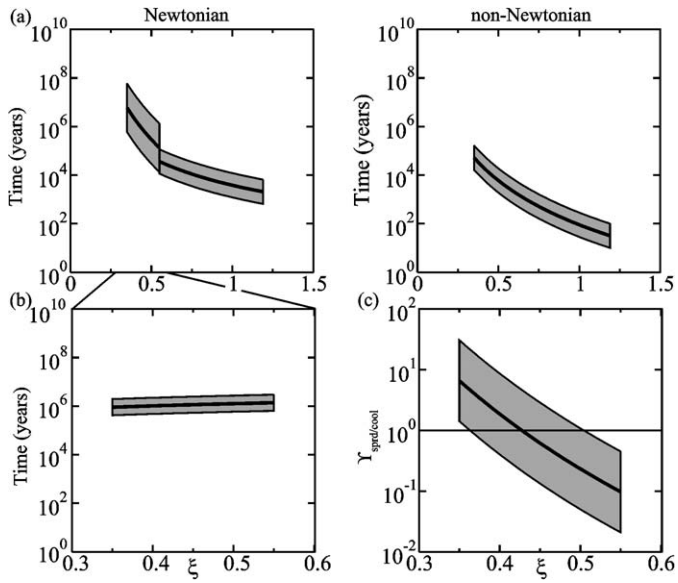


Fig. 10. (a) Timescale for surface spreading of partially molten material as a function of partial melt region size for (left) Newtonian and (right) non-Newtonian radial relaxation. For all $\xi$ in the non-Newtonian case and for $\xi > 0.55$ in the Newtonian case, the spreading is modeled as a plastic gravity current (40), and the spreading timescale is of the same order as the radial relaxation time. There is a transition to a viscous spreading regime for $\xi < 0.55$ in the Newtonian radial relaxation case and the spreading timescale is longer than the radial relaxation time. (b) The bulk cooling time in the viscous spreading regime as a function of partial melt region size. For sufficiently small $\xi$, the cooling time can be shorter than the lateral spreading time. (c) The ratio of the spreading time to the bulk cooling time in the viscous spreading regime as a function of partial melt region size. For $\xi < 0.45$, bulk cooling brings spreading to a halt before isostatic adjustment is complete.

bulk cooling can be characterized by the ratio of the spreading and cooling timescales

$$\Upsilon_{\text{sprd/cool}} \sim t_{\text{sprd}}/t_{\text{cool}}. \tag{42}$$

For $\Upsilon_{\text{sprd/cool}} \ll 1$, the partial melt region is rapidly redistributed and spherical symmetry is restored. For $\Upsilon_{\text{sprd/cool}} \gg 1$, the partial melt cools and lateral spreading stops before a global layer is formed. The value of $\Upsilon_{\text{sprd/cool}}$ depends on the melt region size [Fig. 10(c)]. For $\xi < 0.45$, bulk cooling can bring the spreading to a halt before isostatic adjustment is complete.

The validity of this scaling can be evaluated by comparing basal and crustal viscous stress. Basal viscous stress dominates throughout spreading if $\eta_c/\eta_m \ll (2R_p)^{-3}(\rho_{cr} g V^3)/(\kappa\eta_m)^{1/2}$. Thus, for a local magma ocean near the transition, the crustal viscous force can play a role if $\eta_c > 10^{20}$ Pa s, which is a possibility. However, in this case, the crustal viscous stress can be large because the crustal to basal area ratio is small. In a crust controlled spreading regime, yielding occurs for $\eta_c < V^5\rho_{cr}^2 g^2/(R_p^{11}\kappa\tau_y)$. At the transitional value for $\xi$, this condition is $\eta_c < 30$ Pa s. Thus, there is likely no value of $\eta_c$ for which the crust can play a dynamical role for $\xi < 0.55$.

### 4.4. Comparison of timescales for partially molten local magma ocean evolution

Convective cooling of the partial melt pond beyond the critical crystal fraction via thermally activated creep takes $\sim 10^8$–$10^9$ yr depending on the efficiency of surface recycling (Fig. 7, Table 2). The compaction timescale is very short and melt migration in the partially molten magma ocean is controlled by percolation. Differentiation takes $\sim 10^8$ yr (Fig. 8, Table 2). Isostatic adjustment is the fastest process and occurs prior to any cooling or differentiation. The timescale for isostatic adjustment is $\sim 10^2$–$10^5$ yr for nominal solid mantle rheological parameters and plastic deformation of partially molten silicates (Fig. 10, Table 2). For smaller local magma ocean size ($\xi < 0.55$) and Newtonian radial relaxation, isostatic adjustment is limited by lateral spreading in a viscous regime and can be much longer $\sim 10^5$–$10^7$ yr. In this case, the spreading time can be longer than the bulk cooling time and isostatic adjustment can cease before a globally symmetric mass distribution is restored (Fig. 10).

## 5. Discussion

Large impacts during late stage Mars formation are an inevitable consequence of terrestrial planet formation and can produce large, intact, local melt ponds. The results of this study suggest two end-member cases for the subsequent evolution of such a local magma ocean. (1) If the solid part of the planet has a sufficiently weak rheology and convective cooling and crystallization of the local magma ocean is inefficient, rapid isostatic adjustment results in redistribution of the local magma ocean in a mostly liquid state. A global melt layer of uniform depth, i.e., a global magma ocean is formed. The ocean depth is $\sim \frac{1}{6}R_p\xi(1 - 3/8\xi) \sim 300$ km for $\xi \sim 1$. The solid planet rheology depends primarily on the temperature distribution prior to the melt pond forming impact and thus is sensitive to accretional heating prior to large impacts and also the potential for superposition of several large impacts. (2) For strong solid planet rheology and rapid crystallization, the melt region crystallizes in situ to a partially molten state. Crystallization is associated with a rheological transition from the low viscosity of liquid silicates to the high viscosity of partially molten solids. Isostatic adjustment of the partial melt region is fast compared to any further convective cooling or melt migration and differentiation. For large local magma oceans, large stresses result in plastic deformation and rapid lateral spreading compared to the bulk cooling timescale. At the end of spreading the mass distribution is not symmetric but would continue to relax viscously with a timescale that would depend on the effective bulk viscosity. For small local magma oceans and longer radial relaxation time (Newtonian solid planet viscosity), there is a transition to a viscous spreading regime. In this case, for sufficiently small $\xi$, the spreading time can be longer than the bulk cooling time and spreading would stop before isostatic adjustment was complete (Fig. 11).

### 5.1. A hypothesis for the origin of the hemispherical crustal dichotomy and Tharsis

The martian crustal hemispherical dichotomy is expressed topographically as a north to south slope of $\sim 0.036°$ such that the south pole is $\sim 6$ km higher than the north pole (Smith et al., 1999). The boundary zone topography is complex (Frey

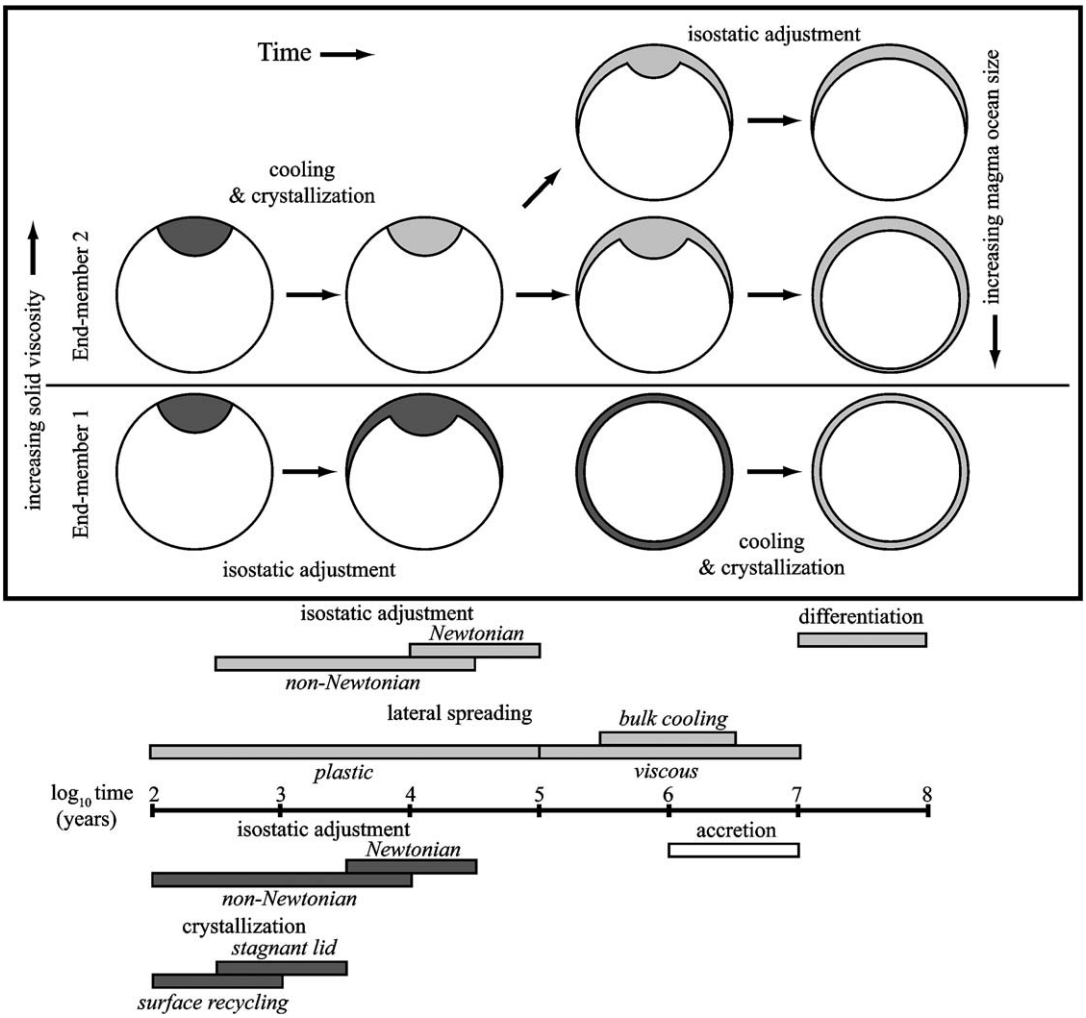


Fig. 11. (Bottom) Characteristic fluid dynamical timescales for a local martian magma ocean. Ranges correspond to variation due to rheological uncertainties and local magma ocean size. The dark and light shading correspond to liquid and partially molten regions, respectively. Crystallization timescales are labeled by the convective regime (surface recycling and stagnant lid). Isostatic adjustment times of the melt region are limited by radial relaxation and are given for both Newtonian and power-law solid planet viscosity. Convective cooling times of the partial melt region are very long ∼300 Myr–3 Gyr and not shown here. (Top) The evolution of a local magma ocean depends on the effective solid planet viscosity, the mechanism of melt region crystallization and the melt region size. Time increases to the right. For end-member 1, isostatic adjustment occurs prior to crystallization resulting in a global layer of uniform depth. For end-member 2, crystallization to a partially molten state occurs prior to isostatic adjustment. In this case for larger melt region size, lateral spreading in a plastic regime results in an asymmetric global layer. Further relaxation depends on the mean thickness and viscosity structure. For smaller melt regions, lateral spreading in a viscous regime can result in cessation of mass redistribution prior to global layer formation.

et al., 1998). Geologically, the dichotomy is apparent from the contrast between heavily cratered southern highlands and smoother northern lowlands. The size–frequency distribution of circular depressions in the resurfaced northern plains is similar to the crater distribution in the south implying that southern basement crust is of approximately the same age as northern crust, i.e., early Noachian (Frey et al., 2002). This is consistent with geochemical constraints on differentiation. Mars crustal structure is constrained by topography and gravity data (Smith et al., 2001; Lemoine et al., 2001) from Mars Global Surveyor (MGS). Models with uniform crustal density and varying crustal thickness (Zuber et al., 2000; Zuber, 2001; Neumann et al., 2004) consistent with the data predict a lower bound on the mean crustal thickness of ∼50 km and a general crustal thinning trend from south to north. The crustal structure in the dichotomy boundary zone is complex exhibiting a smooth transition in Arabia Terra but more scarp-like regional offsets elsewhere.

Several hypotheses have been suggested for hemispherical dichotomy formation. Excavation by one or more large, basin-forming impacts (Wilhelms and Squyres, 1984; Frey and Schultz, 1988) is inconsistent with the lack of correlation between crustal structure and the boundary surface expression, non-circular shape and complex boundary topography, and lack of filled basin gravity signatures such as Utopia for the northern lowlands as a whole (Zuber et al., 2000; Zuber, 2001). The late stages of a transient episode of plate tectonics has been suggested (Sleep, 1994) although subduction of 35 km thick crust (mean lowland thickness) may be difficult due to the deeper basalt–eclogite transition on Mars. Long wavelength (spherical harmonic degree 1) mantle convection is possible for Mars but requires a layered viscosity

structure (Zhong and Zuber, 2001). Such a convective planform could produce a dichotomy by enhanced melting and crustal production at the upwelling or crustal erosion by the downwelling. Timescales for surface recycling or mantle flow are difficult to reconcile with early formation of the dichotomy (Solomon et al., 2005) and preservation of Utopia crustal structure makes northern crustal thinning unlikely (Neumann et al., 2004). Gravitational instability due to crystallization of a global magma ocean can lead to mantle overturn that may have a strong degree 1 component and might provide a mechanism for dichotomy formation (Hess and Parmentier, 2001; Elkins-Tanton et al., 2005).

Later in the Noachian, magmatism and tectonism were focused at Tharsis province, a broad topographic rise comprised of regional centers of volcanism and deformation (e.g., Anderson et al., 2001). Large crustal structure variations (Zuber et al., 2000) and layered volcanics in Valles Marineris (McEwen et al., 1999) support the notion of constructional magmatism as the primary contributor to Tharsis elevation (Solomon and Head, 1982). Localized magmatism resulted in a lithospheric load with an estimated volume of $\sim 3 \times 10^8$ km$^3$ (Phillips et al., 2001). Faulting patterns dating to the late Noachian are consistent with a flexural loading model based on present-day topography and gravity suggesting that the load was emplaced by that time (Banerdt and Golombek, 2000). Volcanism at Tharsis continued throughout martian history but at much lower levels (Hartmann and Neukum, 2001). Tharsis formation probably postdates development of the hemispherical crustal dichotomy. Crustal thickening is superimposed on the global south to north trend (Zuber et al., 2000; Zuber, 2001; Neumann et al., 2004) and Tharsis volcanic units are emplaced on older magnetized crust (Johnson and Phillips, 2005).

Hypotheses for Tharsis formation have focused on a planform of mantle convection dominated by a single thermal plume as an explanation of the monopolar magmatic and tectonic activity. Deep mantle mineralogical phase changes are capable of stabilizing a one-plume pattern of convection (Weinstein, 1995; Harder and Christensen, 1996; Breuer et al., 1998; Spohn et al., 1998; Harder, 2000) but no plume model has reproduced Tharsis development on a timescale consistent with observation. Alternatively, if the martian mantle is layered and thick southern crust acts as an insulating layer, hot plumes can develop beneath the insulating layer (Wenzel et al., 2004) although this hypothesis does not account for Tharsis' location near the dichotomy boundary. The spatial correlation of the Tharsis rise and dichotomy boundary motivates the hypothesis of edge driven convection as mechanism for Tharsis formation (King and Redmond, 2005). This hypothesis requires a deep, cold, stable, layer beneath southern hemisphere crust in order to initiate small scale convection near the dichotomy boundary.

The results of this study of local magma ocean evolution suggest another hypothesis for formation of the hemispherical dichotomy and Tharsis. A relatively large, impact induced, local magma ocean rapidly crystallizes and isostatic adjustment produces a global layer of partially molten silicates which exhibits a degree-1 asymmetry (Section 4, Figs. 11 and 12). A second smaller impact occurs after isostatic adjustment to the first

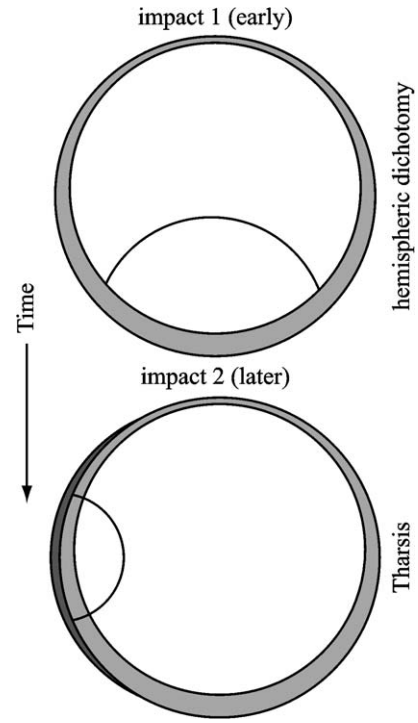


Fig. 12. A diagrammatic version of the hypothesis for formation of the hemispherical crustal dichotomy and Tharsis. The dichotomy is associated with an early Noachian local magma ocean which quickly ($\sim 10^4$–$10^5$) yr evolves to a partially molten asymmetric global layer. During the middle-late Noachian, a smaller local magma ocean is formed which evolves to an axisymmetric dome which is associated with the Tharsis rise. Variable crustal structure within Tharsis may be due to the merger of late stage magma ocean evolution with mantle dynamics which can lead to more localized melting (Reese et al., 2004).

partially molten magma ocean is complete. This local magma ocean crystallizes and its size is such that the lateral spreading proceeds in a viscous regime (Fig. 10). In this case, bulk cooling results in cessation of spreading and formation of a large axisymmetric dome of material (Figs. 11 and 12). In this scenario, both the dichotomy and Tharsis are the result of similar processes of local magma ocean evolution the only difference being the size of the initial, impact induced melt regions.

In the context of this hypothesis, the earliest formed martian crust is simply the solidified partial melt layer produced by the larger of the two impacts. A mean crustal thickness of 50 km, corresponding to 4.4% of the planetary volume Zuber (2001), requires a local magma ocean size of $\xi \sim 0.5$ [Eq. (5)]. In this case, the lateral component of isostatic adjustment occurs in a plastic deformation regime and takes $\sim 10^4$–$10^5$ yr. The layer thickness decreases from $Z \sim (4\tau_y R_p/(\rho_{cr} g))^{1/2} \sim 100$ km on axis to near zero at the antipode (Fig. 12). Of course, while this layer continues to cool and solidify it also can relax viscously. Long-term preservation of the thickness variation depends on the mean thickness and viscosity structure (Zuber et al., 2000; Nimmo and Stevenson, 2001).

Formation of Tharsis can be due to the evolution of a smaller impact induced melt region. At the low end of the range of $\xi$, the retained, complete melt volume is $\sim 5 \times 10^{17}$ m$^3$ which is of the order of the volume estimated for magmatic material comprising Tharsis (Phillips et al., 2001). The distribution

of this material after isostatic adjustment can be estimated from the viscous lateral spreading model (41). The cooling time is $\sim 10^6$ yr (Fig. 10), the radius $S \sim 3000$ km corresponding to an area of $\sim 25\%$ of the planetary surface area [Eq. (41)], and the thickness is $Z \sim (\eta_m V/(\rho_{cr} g))^{1/4} t_{\text{cool}}^{-1/4} \sim 17$ km. Large crustal thickness variability associated with Tharsis (Zuber et al., 2000; Zuber, 2001) may be related to regionally enhanced melting due to longer timescale mantle dynamics induced by late stage, local magma ocean evolution (Reese et al., 2004).

### 5.2. Unanswered questions and future work

The results of this study raise several questions regarding the formation and early evolution of Mars. (i) A critical factor in the models is the effective, solid-planet viscosity which is strongly temperature dependent (Section A.1). Further studies of Mars thermal evolution during accretion which, e.g., include planetesimal heating would help to constrain the early temperature profile. (ii) If the impact interval time is shorter than the individual local magma ocean evolution time, superposition of several large impacts might result in development of a global, laterally heterogeneous, magma ocean. This possibility is intriguing in light of geochemical evidence for distinct mantle reservoirs that have survived since early in martian evolution (Albarède et al., 2000; Halliday et al., 2001; Borg et al., 2003; McLennan, 2003; Kleine et al., 2004; Foley et al., 2005). Furthermore, the possibility exists that compositional differentiation might play a dynamical role in local magma ocean evolution. If melt transport is sufficiently fast, the differentiation timescale can be small and compositional convection might be important (e.g., Elkins-Tanton et al., 2003). Section 4.2.2 suggests that the melt transport mechanism must be faster than percolation for differentiation to play a dynamical role. (iii) The insulating effects of an impact induced atmosphere were not included in the study but can play a role in the fluid dynamics of local magma ocean evolution (Matsui and Abe, 1986; Zahnle et al., 1988). (iv) Detection of magnetized ancient crust (Acuña et al., 1999; Connerney et al., 1999) imply that Mars had a transient core dynamo early in planetary history (Stevenson, 2001). While early mantle cooling (Nimmo and Stevenson, 2000) and an initially hot core (Williams and Nimmo, 2004) have been studied, convection in a magma ocean also marginally satisfies the criterion for generation of a dynamo (Stevenson, 2001). (v) Finally, the initial evolution of a local magma ocean can eventually merge with the subsequent mantle dynamics. This question was addressed in a preliminary study of the effects of initial thermochemical buoyancy associated with a local magma ocean on the early magmatic evolution of Mars (Reese et al., 2004).

## Appendix A. Rheology of solid, partially molten, and liquid silicates

### A.1. Solid silicate viscosity

In steady-state ductile creep, the viscosity law is

$$\eta = \frac{1}{2A}\mu^n \left(\frac{h}{b^*}\right)^m \tau^{1-n} \exp\left(\frac{Q}{RT}\right), \tag{A.1}$$

where $A$ is the pre-exponential factor, $\mu$ is the shear modulus, $n$ is the power law exponent, $h$ is the grain size, $b^*$ is the Burger vector length, $\tau$ is the second invariant of the deviatoric stress tensor, $T$ is the temperature, $R$ is the gas constant, and $Q = E + pV$ is the activation enthalpy, $E$ is the activation energy, $p$ is the pressure, $V$ is the activation volume, $n = 1$ for diffusion creep, and $n = 3.0$–$3.5$ for dislocation creep (Karato and Wu, 1993).

The effective viscosity controlling large scale deformation is estimated from the average mantle value along an adiabat,

$$\eta_s \approx \frac{\mu^n}{2A}\left(\frac{h}{b^*}\right)^m \tau^{1-n} \exp\left(\frac{E}{RT_0}\right)\left[\frac{p_{\text{cmb}}}{p_T}\left(\frac{p_T V - E}{RT_0}\right)\right]^{-1} \times \left[\exp\left[\frac{p_{\text{cmb}}}{p_T}\left(\frac{p_T V - E}{RT_0}\right)\right] - 1\right], \tag{A.2}$$

where $T_0$ is the potential temperature, $p_{\text{cmb}}$ is the pressure at the core mantle boundary, $p_T = \rho c_p/\alpha$ is the scale pressure, and terms of the order $(p^2/p_T^2)$ and higher have been dropped.

For potential temperature $T_0 \sim 1400$ K,

$$\eta_s \sim \begin{cases} 1 \times 10^{21}\text{–}3 \times 10^{23}\ \text{Pa s}(h/1\ \text{mm})^{2.5}, & n = 1, \\ 1 \times 10^{35}\text{–}1 \times 10^{44}\ \text{Pa}^n\ \text{s}\,\tau^{1-n}, & n = 3. \end{cases} \tag{A.3}$$

For diffusion creep it is assumed that $\eta_s \sim 10^{21}$ Pa s which is close to the present-day terrestrial upper mantle viscosity. For dislocation creep, $\eta_s \sim b_s \tau^{1-n}$ with $b_s \sim 10^{37}$ Pa$^n$ s and a large uncertainty of several orders of magnitude.

### A.2. Partially molten silicate viscosity

The presence of melt results in an enhanced creep rate. Recent experiments (Hirth and Kohlstedt, 1995a, 1995b; Kohlstedt and Zimmerman, 1996; Hirth and Kohlstedt, 2003; Scott and Kohlstedt, 2004a, 2004b) suggest that at melt fractions $\phi > 0.05$, the viscosity law

$$\eta = \frac{1}{2A}\mu^n \left(\frac{h}{b^*}\right)^m \tau^{1-n} \exp(-\alpha_\eta \phi) \exp\left(\frac{Q}{RT}\right), \tag{A.4}$$

where $\alpha_\eta \sim 25$ and 30 for diffusion and dislocation creep, respectively (Scott and Kohlstedt, 2004b). At larger melt fractions, $\phi > 0.25$, a several order magnitude decrease in viscosity over a narrow melt fraction range (Scott and Kohlstedt, 2004b) is consistent with a rheological transition. For the purposes of this study, the partially molten mantle viscosity is defined at $\phi_m = 0.2$.

The solidus and liquidus are parameterized (Reese et al., 1999) based on data for peridotites (McKenzie and Bickle, 1988; Scarfe and Takahashi, 1986; Ito and Takahashi, 1987; Herzberg and Zhang, 1994),

$$T_s = \begin{cases} a_1 + a_2 p - a_3 p^2, & p < p_{tr}, \\ b_1 + b_2(p - p_{tr}), & p > p_{tr}, \end{cases} \tag{A.5}$$

and

$$T_l = c_1 + c_2 p + c_3 \tan^{-1}\left(\frac{p}{c_4}\right), \tag{A.6}$$

where $a_1 = 1374$ K, $a_2 = 130$ K/GPa, $a_3 = 5.6$ K/GPa$^2$, $b_1 = 2124$ K, $b_2 = 10$ K/GPa, and $p_{tr} = 10.7$ GPa, $c_1 = 2009$ K,

Table A.1
Rheological parameters

| | Diffusion creep | | Dislocation creep | |
|---|---|---|---|---|
| | Wet | Dry | Wet | Dry |
| $A$ ($s^{-1}$) | $5.3 \times 10^{15}$ | $8.7 \times 10^{15}$ | $2.0 \times 10^{18}$ | $3.5 \times 10^{22}$ |
| $n$ | 1.0 | 1.0 | 3.0 | 3.5 |
| $m$ | 2.5 | 2.5 | 0.0 | 0.0 |
| $E$ (kJ/mol) | 240 | 300 | 430 | 540 |
| $V$ ($cm^3$/mol) | 5 | 6 | 10–20 | 15–25 |

$c_2 = 4.3$ K/GPa, $c_3 = 180$ K, $c_4 = 2.2$ GPa. For linear melt fraction variation between $T_s$ and $T_l$, the temperature corresponding to a melt fraction $\phi_m$ is $T_m = T_s + \phi_m(T_l - T_s)$.

For a partially molten mantle, at a pressure of 1 GPa, the range of viscosities implied by the rheological laws and Table A.1 are

$$\eta_m \sim \begin{cases} 1 \times 10^{17}\text{–}3 \times 10^{18}\ \text{Pa s}(h/1\ \text{mm})^{2.5}, & n = 1, \\ 1 \times 10^{27}\text{–}1 \times 10^{32}\ \text{Pa}^n\ \text{s}\,\tau^{1-n}, & n \approx 3. \end{cases} \tag{A.7}$$

For $n = 1$ and $n \approx 3$, it is assumed that $\eta_m \sim 10^{17}$ Pa s and $b_m \sim 10^{30}$ $\text{Pa}^n$ s, respectively.

### *A.3. Liquid silicate viscosity, rheological transition, and viscosity of partial melt near the solidus*

For ultramafic silicates at low pressure and large melt fraction, the viscosity $\eta_l \sim 0.1$ Pa s (Bottinga and Weill, 1972; Shaw, 1972; Persikov et al., 1990; Bottinga et al., 1995). Variations due to pressure are small for a completely depolymerized melt (Andrade, 1952; Gans, 1972). Thus, the viscosity of the magma ocean near the liquidus is assumed to be $\eta_l \sim 0.1$ Pa s.

Crystal fraction is the primary parameter affecting the viscosity of partially molten silicates. Models of concentrated suspensions (Mooney, 1951; Roscoe, 1952; Brinkman, 1952; Krieger and Dougherty, 1959; Murray, 1965; Frankel and Acrivos, 1967; Campbell and Forgacs, 1990) and experiments with partial melts (e.g., Arzi, 1978) suggest a rheological transition from a low viscosity crystal suspension to a partially molten solid over a narrow melt fraction range near $\phi = \phi_{cr} \sim 40\%$ (Solomatov and Stevenson, 1993; Solomatov, 2000).

Near the solidus, partial melt viscosity is higher due to lower temperatures. At low pressures, basaltic melt viscosity formed near the anhydrous solidus $\eta_f \sim 100$ Pa s (Kushiro, 1980, 1986) depending on composition and, in particular, silica content which affects degree of polymerization. Viscosities of $10^4$ Pa s have been estimated for basaltic lavas on Mars (Warner and Gregg, 2003) although this value is certainly affected by the presence of phenocrysts and bubbles. A value of $\eta_f \sim 100$ Pa s is assumed.

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